

Lecture 8 ERS312 Notes

- By definition seamounts are extinct volcanoes that have never reached sea level while table mounts have reached sea level and eroded
- Seamounts always been underwater

Plate Tectonics

- What's happening in sea floor (plate tectonics theory first noticed by Alfred Wegener)
- Sonar systems in World War II, largely to find submarines and get ideas of sea floor (submarines travelled close to sea floor)
- Magnetometers to also locate submarines (was pattern of different magnetic orientation on ocean floor)
- These were applied to research after WW2
- Normal polarity magnetic compass needle pointed to north (reversed polarity pointed to south)
- Magnetic reversals are recorded in rock
- Overtime gets a pattern of normal and reversed polarities (can date the rock)
- Noticed same on ocean floor
- Started finding magnetic signals in oceanic crust (basalt)
- When magma (underground) rises it becomes lava
- Due to plate movement solidified lava is carried away from both sides of ridge
- Lead to hypothesis that seafloor is spreading

Age of Oceanic Crust

- Youngest crust is found in center of Atlantic
- Oldest oceanic crust is about 144-200 million years (furthest away from mid ocean ridges)
- Oldest ocean crust dates to Jurassic (200 Ma)
- Why are ocean so young compared to land (they are constantly being created, gets older and older and subduct going under mantle and melts, coming back up)
- Seafloor spreading and subduction are essential components of Plate Tectonics

Distribution of plates

- Pacific plate is an oceanic plate

Wilson Cycle & features of Ocean Crust

- Sometimes continents break apart
- Get a mid ocean ridge forming, lava coming out
- Opening of ocean basin, gets too old and subducts
- Takes 200-600 million years
- As crust gets older there's more sediment on crust
- Thickest sediment cover is close to land
- Thinnest sediment cover is at center of Atlantic (because it's in the ocean ridge) they don't collect on ridge (that's where crust forms, youngest) *for sediment to accumulate you need time
- Young oceanic crust is less dense and more buoyant (highest topographic feature)
- Depth of crust increases with increasing crustal age
- Pacific is spreading much faster than Atlantic
- Fast Spreading ridge (East pacific), Slow spreading (mid-Atlantic ridge), ultra slow (southwest Indian ridge)

Ridge-axis discontinuities

- Ridge is not continuous but is offset *shifted (lateral movement of ridge)
- It is discontinuous
- Laterally moving crusts in opposite directions (fault)
- Exercise: Offset to left because depth difference (top portion is further away from ridge because depth is higher so offset to left)
-average spreading rate: $d/t \times \text{time} = \text{age of crust}$
 $1000\text{km}/50 \text{ Ma} = 2 \text{ cm/yr}$ (this is a slow spreading ridge)

Ridge Axis Processes

- Sea water above sea floor with a lot of pressure
- Water leaching the rock (sea water gets heated and comes out of sea floor, contains a lot of different elements from rock)

